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Automatic well killing system

Abstract

A system includes a wellhead assembly installed above a well, the wellhead assembly including a passageway extending therethrough in fluid communication with the well, a sensor operable to detect a characteristic of a fluid flow in the passageway indicative of a need for a well killing operation, a kill fluid line extending between a source of kill fluid and the passageway, a hydraulic control remote (HCR) valve installed in the kill fluid line, the HCR valve including a gate movable between closed and open positions with respect to the kill fluid line and an actuator to open and close the gate, and a controller communicatively coupled to the flow sensor and the HCR valve operable to instruct the actuator of the HCR valve to open in response to determining the characteristic of the fluid flow in the passageway is indicative of a need for a well kill operation.

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Background/Summary

FIELD OF THE DISCLOSURE

(1) The present disclosure relates generally to well killing operations and, more particularly, to automatic well killing via a modified hydraulic control remote valve and automated blowout preventer.

BACKGROUND OF THE DISCLOSURE

(2) During drilling and hydrocarbon extraction operations in a hydrocarbon-producing well, the need may arise to prevent the flow of formation fluids to the well surface, also known as “killing” of the well. A common well killing operation involves introducing a heavy fluid (known as “kill fluid” or “kill mud”) into the well in order to overcome the pressure of the formation fluids flowing from the subterranean formation and cease formation fluid flow. Well killing may be required

during advanced interventions, such as workovers, or may be performed as contingency operations to prevent uncontrolled flow of the formation fluids to the surface, or blowouts.
(3) Efforts have been made to enable remote killing of wells using a hydraulic control remote (HCR) valve in the choke or kill fluid lines near the wellhead, such that kill fluid may be allowed to flow into the well without an operator manually opening a gate valve. However, manually initiating the well killing operation, even remotely, may place the well, equipment, and personnel at risk during the detection, preparation, and execution phases of the well killing operation.
(4) Accordingly, systems and methods for the automatic detection of problematic flow, and subsequent automatic well killing, are desirable.
SUMMARY OF THE DISCLOSURE
(5) Various details of the present disclosure are hereinafter summarized to provide a basic understanding. This summary is not an exhaustive overview of the disclosure and is neither intended to identify certain elements of the disclosure, nor to delineate the scope thereof. Rather, the primary purpose of this summary is to present some concepts of the disclosure in a simplified form prior to the more detailed description that is presented hereinafter.
(6) According to an embodiment consistent with the present disclosure, a system includes a wellhead assembly installed above a well, the wellhead assembly including a passageway extending therethrough in fluid communication with the well, a sensor operable to detect a characteristic of a fluid flow in the passageway indicative of a need for a well killing operation, a kill fluid line extending between a source of kill fluid and the passageway, a hydraulic control remote (HCR) valve installed in the kill fluid line, the HCR valve including a gate movable between closed and open positions with respect to the kill fluid line and an actuator to open and close the gate, and a controller communicatively coupled to the flow sensor and the HCR valve. The controller is operable to instruct the actuator of the HCR valve to open in response to determining the characteristic of the fluid flow in the passageway is indicative of a need for a well kill operation to permit the kill fluid to enter the well through the passageway through the kill fluid line.
(7) In another embodiment, a method includes detecting, with a sensor, a characteristic of a fluid flow through a passageway defined through a wellhead assembly, receiving, with a controller, a signal from the sensor indicative of the characteristic of the fluid flow, determining, with the controller, whether or not a predetermined set of conditions are met indicating a need for the well killing operation based on the signal, actuating one or more hydraulic control remote (HCR) valves within a kill fluid line extending from the passageway in response to determining that the predetermined set of conditions are met, and activating one or more discharge pumps installed within the kill fluid line to pump a kill fluid into the passageway through the one or more HCR valves, wherein the kill fluid is of a density to overcome a pressure of the fluid flow.
(8) Any combinations of the various embodiments and implementations disclosed herein can be used in a further embodiment, consistent with the disclosure. These and other aspects and features can be appreciated from the following description of certain embodiments presented herein in accordance with the disclosure and the accompanying drawings and claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a schematic view of an example system for automatic well killing operations, according to at least one embodiment of the present disclosure.
(2) FIGS. **2-4** are partial cross-sectional views of the system of FIG. **1** in progressive stages of an example automatic well killing operation.
(3) FIG. **5** is a schematic diagram of an example of a computer system that can be employed to

execute one or more embodiments of the present disclosure.

DETAILED DESCRIPTION

(4) Embodiments of the present disclosure will now be described in detail with reference to the accompanying Figures. Like elements in the various figures may be denoted by like reference numerals for consistency. Further, in the following detailed description of embodiments of the present disclosure, numerous specific details are set forth in order to provide a more thorough understanding of the claimed subject matter. However, it will be apparent to one of ordinary skill in the art that the embodiments disclosed herein may be practiced without these specific details. In other instances, well-known features have not been described in detail to avoid unnecessarily complicating the description. Additionally, it will be apparent to one of ordinary skill in the art that the scale of the elements presented in the accompanying Figures may vary without departing from the scope of the present disclosure.

(5) Embodiments in accordance with the present disclosure generally relate to well killing operations and, more particularly, to automatic well killing via a modified hydraulic control remote valve and an automated blowout preventer. The systems and methods described herein may enable automatic detection of problematic flow and an automatic initiation of well killing operations, such that a future blowout may be detected and resolved prior to the development of any danger to operators or damage to equipment. The systems and method described herein may be further installed or implemented alongside standard wellhead and blowout preventer equipment, such that the well killing automation may be retrofit into existing well systems.

(6) FIG. **1** is a schematic view of an example system **100** for automatic well killing operations, according to at least one embodiment of the present disclosure. The system **100** includes a wellhead assembly **101**, which may be installed above a well (not shown) extending through a subterranean formation. The wellhead assembly **101** may include a blowout preventer **102** (partially shown here) mated to a wellhead **104** via a spool **106**. A passageway **110** may be defined through the wellhead assembly **101**, which may extend through each of the blowout preventer **102**, the spool **106**, and the wellhead **104**. The passageway **110** may be in fluid communication with the subterranean formation such that formation fluids may pass through the passageway **110**. As described in greater detail below, the blowout preventer **102** and the fluids introduced through the spool **106** may facilitate the prevention of problematic flows of production fluids through the passageway **110**.

(7) In the illustrated embodiment, the passageway **110** includes piping **111** which may be production tubing, drill string, or other piping which may be run downhole through the wellhead assembly **101**. One or more rams **108**, either blind rams, pipe rams or shear rams, of the blowout preventer **102** may abut the passageway **110**. The one or more rams **108** may be actuated prior to a well killing operation such that further flow of formation fluids to the surface may be obstructed or sealed through blockage or shearing of the passageway **110** or the piping **111**.

(8) The spool **106** may additionally contain internal components such that the spool **106** is a casing head spool or a tubing head spool for further operations. The spool **106** may be mated to, or otherwise in fluid communication with, a mud cross **112** of the wellhead assembly **101**. The mud cross **112** is formed of a series of valves to control a choke line **112***a* and kill fluid lines **112***b* extending from the passageway **110**. One or more of the valves of the mud cross **112** may be the modified hydraulic control remote (HCR) valves **114**, wherein an internal gate may be raised into the valve bonnet remotely in response to a control signal, such as an electronic signal or another type of control signal.

(9) The electronic or other control signal may be autonomously provided. To this effect, the system **100** may further include a controller, such as control panel **116**, which may control the operations of the various valves, including the modified HCR valves **114**, and the one or more rams **108**. The control panel **116** may be communicatively coupled to the remainder of the system **100** via one or more physical cables **118**. In at least one embodiment of the present disclosure, however, the control panel **116** may include a wireless communication component **120** which enables the control

panel **116** to be portable in nature. The wireless communication component **120** may communicate with the remainder of the system **100** via Wifi, bluetooth, infrared, radio frequency, or any other available wireless communication technology.

(10) The system **100** may include a kill tank **122** in fluid communication with the wellhead assembly **101**, such that a kill fluid **124** may be mixed and prepared prior to an active well killing operation. As illustrated, the kill tank **122** may be open such that the kill fluid **124** may be produced, tested, and modified. The kill fluid **124**, once prepared, may be pumped through a first discharge pump **126***a*, such that the kill fluid **124** is directed into an isolated kill tank **128**. The isolated kill tank **128** may be closed and sealed such that the kill fluid **124** may be stored, maintained, and pressurized in preparation for a well killing operation. During a well killing operation, the kill fluid **124** stored in the isolated kill tank **128** may be pumped to the mud cross **112** using a second discharge pump **126***b*. Following the opening of the modified HCR valves **114**, the kill fluid **124** may be injected into the passageway **110** and may flow into the wellhead **104**. The density of the kill fluid **124** allows the kill fluid **128** to overcome the production pressure of the formation fluid, such that the flow of formation fluid to the well surface ceases and the well is effectively killed.

(11) The fluid connections between the kill tank **122**, the isolated kill tank **128**, and the mud cross **112** may include one or more one-way valves **130** interposed within the fluid connections. The one-way valves **130** may prevent backflow of formation fluids into the isolated kill tank **128** or the kill tank **122**, while allowing the kill fluid **124** to be pumped to the mud cross **112**.

(12) The system **100** may enable near-instantaneous killing of the well during blowout events, prior to the presence of danger or damage, and may be fully automated to overcome human error. In some embodiments, the system **100** may be employed in a well killing operation prior to advanced workover, such that the system **100** is manually triggered but automatically performs the well killing operation. The system **100** may be triggered by the control panel **116**, or a remote control device, such that an operator may relocate to a safe distance prior to initiating the well killing operation. The system **100** may thus increase safety associated with both planned and unplanned well killing operations, while adding a layer of automatic blowout prevention to prevent equipment damage and operator harm.

(13) Example operation of the system **100** will now be described with reference to FIGS. **2-4**, which depict progressive cross-sectional views before and during an automatic well killing operation in the wellhead assembly **101**. In FIG. **2**, formation fluid **202** is flowing through the passageway **110**. The presence and flowrate of the formation fluid **202** may be detected by a sensor **204**, which may be installed within the blowout preventer **102** as illustrated or at other locations within or abutting the passageway **110**. In some embodiments, the sensor **204** is an ultrasonic sensor clamped onto an outer wall of the passageway **110** or an outer wall of the piping **111** of FIG. **1**. In these embodiments, the sensor **204** may detect a flowrate or quantity of formation fluid **202** entering from below. The quantity, flowrate, or rate of change of flowrate for the formation fluid **202** may be compared against threshold values to determine the need for a killing operation. In alternate embodiments, the sensor **204** may be a mechanical sensor, an electromagnetic sensor, or any other device which may detect abnormal flow within the passageway **110**, abnormal flow within the piping **111** of FIG. **1**, or conditions indicative of the need for a killing operation.

(14) The sensor **204** may communicate with a processor **206** with a signal regarding the flow of formation fluid **202**. In some embodiments, the processor **206**, as illustrated, may be a component of the control panel **116**, such that device control is further implemented by the control panel **116**. In alternate embodiments, however, the processor **206** may be part of a standalone device or incorporated into any automated device within the system **100** of FIG. **1**. The processor **206** may be installed with, or otherwise communicatively coupled to, a memory **208** on which an algorithm, or instructions, are stored for the processor **206** to perform (execute). The memory **208** may further include a database of the sensor readings and data associated with actions taken during the well

killing operations.
(15) The processor **206** may receive a reading or a flagging signal from the sensor **204**, and may subsequently determine whether the flow of the formation fluid **202** requires a well killing operation. The processor **206** may evaluate the reading or flagging signal to determine whether or not a predetermined set of conditions are met indicating a need for the well killing operation. In a non-limiting example, the processor **206** may determine that a well killing operation is necessary when more than five barrels of formation fluid **202** have been detected by the sensor **204**. In some embodiments, the processor **206** may have a pre-loaded algorithm or module for the determination of, and performance of, a well killing operation. As illustrated, the processor **206** may be in communication with the sensor **204**, the modified HCR valve **114**, and the blowout preventer **102** (or the one or more rams **108**) for facilitating the well killing operation.
(16) In FIG. **3**, the processor **206** has instructed the various components of the system **100** of FIG. **1** to begin the well killing operation. Accordingly, the blowout preventer **102** and, more particularly, the one or more rams **108** have been activated to seal off flow through a top of the passageway **110**, or the piping **111** of FIG. **1**. The one or more rams **108** may be actuated, such that the internal rams **302** begin travelling inwards toward the passageway **110**. In some embodiments, the one or more rams **108** may be hydraulically actuated to overcome internal pressures or materials interspersed between the internal rams **302**. The internal rams **302** may continue to travel until the passageway **110** is completely cut, sealed, or compressed. In some embodiments, the one or more rams **108** may be shear rams, such that any pipe or tubing forming, or disposed within, the passageway **110** is sheared and the ram heads **304** may mate to form a seal over the damaged passageway **110**. In alternate embodiments, the one or more rams **108** may be blind rams, such that the ram heads **304** may meet within the passageway **110** and form a plug or cover within the passageway **110** and prevents further flow. In other embodiments, the one or more rams **108** may be pipe rams, such that the ram heads **304** form an annular seal around a drill pipe, production tubing, coiled tubing, or other tubular (e.g., the tubing **111** of FIG. **1**) extending through the flow path **110**. Those skilled in the art will readily appreciate that other forms of rams may be utilized in the scaling of the passageway **110** within the blowout preventer **102** such that further flow of formation fluid **202** is prevented, without departing from the scope of this disclosure.
(17) In FIG. **4**, the internal rams **302** have fully travelled such that the ram heads **304** have created a blockage within the blowout preventer **102**, thus preventing further flow of the formation fluid **202** of FIGS. **2-3**. In some embodiments, the one or more rams **108** may transmit a closure signal to the processor **206** signifying a completed seal. In further embodiments, the sensor **204** may transmit a closure signal to the processor **206** signaling a drop in, or cessation of, flow past the sensor **204** to indicate that the passageway **111** has been completely sealed. The processor **206** may delay instructing the HCR valve **114** until the closure signal has been received. Once the processor **206** has received confirmation of the sealing within the blowout preventer **102**, or has paused execution long enough after signaling for actuation of the one or more rams **108**, the modified HCR valve **114** may be instructed to open.
(18) A hydraulic actuation mechanism **402** of the HCR valve **114** may be instructed, or directly controlled, by the processor **206** to begin upward travel of the gate **404** into the bonnet of the modified HCR valve **114**. As the gate **404** is retracted into the bonnet of the modified HCR valve **114**, the kill fluid **124** may be pumped through the kill fluid line **112***b* of the mud cross **112** and into the passageway **110**. The kill fluid **124** may begin to mix with, or displace the remaining formation fluid **202** of FIGS. **2-3**, and may begin to travel down through the wellhead **104**. The kill fluid **124** may flow through the passageway **110** until reaching the subterranean formation, or may flow until a sufficient hydrostatic head is formed above the formation fluid **202** of FIGS. **2-3**. The formed hydrostatic head may be heavy or dense enough to overcome the upward pressure from the subterranean formation below, thus preventing further upward flow from the subterranean formation, thus killing the well.

(19) The operations described in the discussion of FIGS. **2-4** may be fully automated in operation, such that the detection of formation fluid **202** of FIGS. **2-3**, the sealing of the passageway **110** via the one or more rams **108**, the opening of the modified HCR valve **114**, and the pumping of kill fluid **124** into the well are all automatically initiated and performed. The automation of the well killing operations may enable near-instantaneous resolution of blowout events before damage or danger may occur, and may ensure all proper steps are taken in the well killing process. The automated process may eliminate human error in the well killing operation, but may also enable operators to reach a safe distance from the well during the possibly hazardous event.
(20) In view of the foregoing structural and functional description, those skilled in the art will appreciate that portions of the embodiments may be embodied as a method, data processing system, or computer program product. Accordingly, these portions of the present embodiments may take the form of an entirely hardware embodiment, an entirely software embodiment, or an embodiment combining software and hardware, such as shown and described with respect to the computer system of FIG. **5**. Furthermore, portions of the embodiments may be a computer program product on a computer-readable storage medium having computer readable program code on the medium. Any non-transitory, tangible storage media possessing structure may be utilized including, but not limited to, static and dynamic storage devices, volatile and non-volatile memories, hard disks, optical storage devices, and magnetic storage devices, but excludes any medium that is not eligible for patent protection under 35 U.S.C. § 101 (such as a propagating electrical or electromagnetic signals per se). As an example and not by way of limitation, computer-readable storage media may include a semiconductor-based circuit or device or other IC (such, as for example, a field-programmable gate array (FPGA) or an ASIC), a hard disk, an HDD, a hybrid hard drive (HHD), an optical disc, an optical disc drive (ODD), a magneto-optical disc, a magneto-optical drive, a floppy disk, a floppy disk drive (FDD), magnetic tape, a holographic storage medium, a solid-state drive (SSD), a RAM-drive, a SECURE DIGITAL card, a SECURE DIGITAL drive, or another suitable computer-readable storage medium or a combination of two or more of these, where appropriate. A computer-readable non-transitory storage medium may be volatile, nonvolatile, or a combination of volatile and non-volatile, as appropriate.
(21) Certain embodiments have also been described herein with reference to block illustrations of methods, systems, and computer program products. It will be understood that blocks and/or combinations of blocks in the illustrations, as well as methods or steps or acts or processes described herein, can be implemented by a computer program comprising a routine of set instructions stored in a machine-readable storage medium as described herein. These instructions may be provided to one or more processors of a general purpose computer, special purpose computer, or other programmable data processing apparatus (or a combination of devices and circuits) to produce a machine, such that the instructions of the machine, when executed by the processor, implement the functions specified in the block or blocks, or in the acts, steps, methods and processes described herein.
(22) These processor-executable instructions may also be stored in computer-readable memory that can direct a computer or other programmable data processing apparatus to function in a particular manner, such that the instructions stored in the computer-readable memory result in an article of manufacture including instructions which implement the function specified. The computer program instructions may also be loaded onto a computer or other programmable data processing apparatus to cause a series of operational steps to be performed on the computer or other programmable apparatus to realize a computer implemented process such that the instructions which execute on the computer or other programmable apparatus provide steps for implementing the functions specified in flowchart blocks that may be described herein.
(23) In this regard, FIG. **5** illustrates one example of a computer system **500** that can be employed to execute one or more embodiments of the present disclosure. Computer system **500** can be implemented on one or more general purpose networked computer systems, embedded computer

systems, routers, switches, server devices, client devices, various intermediate devices/nodes or standalone computer systems. Additionally, computer system **500** can be implemented on various mobile clients such as, for example, a personal digital assistant (PDA), laptop computer, pager, and the like, provided it includes sufficient processing capabilities.

(24) Computer system **500** includes processing unit **502**, system memory **504**, and system bus **506** that couples various system components, including the system memory **504**, to processing unit **502**. System memory **504** can include volatile (e.g. RAM, DRAM, SDRAM, Double Data Rate (DDR) RAM, etc.) and non-volatile (e.g. Flash, NAND, etc.) memory. Dual microprocessors and other multi-processor architectures also can be used as processing unit **502**. System bus **506** may be any of several types of bus structure including a memory bus or memory controller, a peripheral bus, and a local bus using any of a variety of bus architectures. System memory **504** includes read only memory (ROM) **510** and random access memory (RAM) **512**. A basic input/output system (BIOS) **514** can reside in ROM **510** containing the basic routines that help to transfer information among elements within computer system **500**.

(25) Computer system **500** can include a hard disk drive **516**, magnetic disk drive **518**, e.g., to read from or write to removable disk **520**, and an optical disk drive **522**, e.g., for reading CD-ROM disk **524** or to read from or write to other optical media. Hard disk drive **516**, magnetic disk drive **518**, and optical disk drive **522** are connected to system bus **506** by a hard disk drive interface **526**, a magnetic disk drive interface **528**, and an optical drive interface **530**, respectively. The drives and associated computer-readable media provide nonvolatile storage of data, data structures, and computer-executable instructions for computer system **500**. Although the description of computer-readable media above refers to a hard disk, a removable magnetic disk and a CD, other types of media that are readable by a computer, such as magnetic cassettes, flash memory cards, digital video disks and the like, in a variety of forms, may also be used in the operating environment; further, any such media may contain computer-executable instructions for implementing one or more parts of embodiments shown and described herein.

(26) A number of program modules may be stored in drives and ROM **510**, including operating system **532**, one or more application programs **534**, other program modules **536**, and program data **538**. In some examples, the application programs **534** can include detection of the formation fluid **202**, actuation of the one or more rams **108**, opening of the modified HCR valve **114**, or pumping of the kill fluid **124**, and the program data **538** can include the flowrate determined by the sensor **204**, or time since actuation of the one or more rams **108**. The application programs **534** and program data **538** can include functions and methods programmed to automatically detect blowout events and initiate well killing operations without operator intervention, such as shown and described herein.

(27) A user may enter commands and information into computer system **500** through one or more input devices **540**, such as a pointing device (e.g., a mouse, touch screen), keyboard, microphone, joystick, game pad, scanner, and the like. For instance, the user can employ input device **540** to edit or modify flow parameters for automatic detection of a blowout event, or to manually initiate a well killing operation for an advanced workover. These and other input devices **540** are often connected to processing unit **502** through a corresponding port interface **542** that is coupled to the system bus, but may be connected by other interfaces, such as a parallel port, serial port, or universal serial bus (USB). One or more output devices **544** (e.g., display, a monitor, printer, projector, or other type of displaying device) is also connected to system bus **506** via interface **546**, such as a video adapter.

(28) Computer system **500** may operate in a networked environment using logical connections to one or more remote computers, such as remote computer **548**. Remote computer **548** may be a workstation, computer system, router, peer device, or other common network node, and typically includes many or all the elements described relative to computer system **500**. The logical connections, schematically indicated at **550**, can include a local area network (LAN) and/or a wide

area network (WAN), or a combination of these, and can be in a cloud-type architecture, for example configured as private clouds, public clouds, hybrid clouds, and multi-clouds. When used in a LAN networking environment, computer system **500** can be connected to the local network through a network interface or adapter **552**. When used in a WAN networking environment, computer system **500** can include a modem, or can be connected to a communications server on the LAN. The modem, which may be internal or external, can be connected to system bus **506** via an appropriate port interface. In a networked environment, application programs **534** or program data **538** depicted relative to computer system **500**, or portions thereof, may be stored in a remote memory storage device **554**.

(29) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention. As used herein, for example, the singular forms "a," "an," and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms "contains", "containing", "includes", "including," "comprises", and/or "comprising," and variations thereof, when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(30) Terms of orientation used herein are merely for purposes of convention and referencing and are not to be construed as limiting. However, it is recognized these terms could be used with reference to an operator or user. Accordingly, no limitations are implied or to be inferred. In addition, the use of ordinal numbers (e.g., first, second, third, etc.) is for distinction and not counting. For example, the use of "third" does not imply there must be a corresponding "first" or "second." Also, if used herein, the terms "coupled" or "coupled to" or "connected" or "connected to" or "attached" or "attached to" may indicate establishing either a direct or indirect connection, and is not limited to either unless expressly referenced as such.

(31) While the disclosure has described several exemplary embodiments, it will be understood by those skilled in the art that various changes can be made, and equivalents can be substituted for elements thereof, without departing from the spirit and scope of the invention. In addition, many modifications will be appreciated by those skilled in the art to adapt a particular instrument, situation, or material to embodiments of the disclosure without departing from the essential scope thereof. Therefore, it is intended that the invention not be limited to the particular embodiments disclosed, or to the best mode contemplated for carrying out this invention, but that the invention will include all embodiments falling within the scope of the appended claims. Moreover, reference in the appended claims to an apparatus or system or a component of an apparatus or system being adapted to, arranged to, capable of, configured to, enabled to, operable to, or operative to perform a particular function encompasses that apparatus, system, or component, whether or not it or that particular function is activated, turned on, or unlocked, as long as that apparatus, system, or component is so adapted, arranged, capable, configured, enabled, operable, or operative.

Claims

1. A system comprising: a wellhead assembly installed above a well and including a blowout preventer comprising one or more rams, wherein a passageway extends through the wellhead assembly and is in fluid communication with the well; a sensor mounted to an interior wall of the passageway and operable to detect a characteristic of a fluid flow in the passageway indicative of a need for a well killing operation; a kill fluid line extending between a source of kill fluid and the passageway; a hydraulic control remote (HCR) valve installed in the kill fluid line and including a gate movable between closed and open positions with respect to the kill fluid line and an actuation mechanism operable to open and close the gate; and a controller communicatively coupled to the sensor and the actuation mechanism, the controller operable to control the actuation mechanism to

open the gate in response to detection of the characteristic of the fluid flow in the passageway indicative of the need for a well kill operation to thereby permit the kill fluid to enter the well through the kill fluid line and the passageway, wherein the controller communicates with the blowout preventer to actuate the one or more rams, and wherein the controller opens the gate after receiving a closure signal from the blowout preventers indicating actuation of the one or more rams.

2. The system of claim 1, further comprising one or more discharge pumps installed within the kill fluid line, wherein the controller is communicatively coupled to the one or more discharge pumps and operable to instruct activation of the one or more discharge pumps subsequent to the actuation of the one or more rams.

3. The system of claim 2, wherein the controller autonomously and directly controls the actuation mechanism, the blowout preventer, and the one or more discharge pumps in response to detection of the characteristic of the fluid flow in the passageway indicative of the need for a well kill operation.

4. The system of claim 1, wherein the source of kill fluid comprises: a mixing tank arranged adjacent the wellhead assembly for preparing the kill fluid; an isolated kill tank in fluid communication with the mixing tank and fluidly coupled to the kill fluid line, the isolated kill tank receiving the kill fluid from the mixing tank and being closed and sealed such that the kill fluid is stored, maintained, and pressurized in preparation for the well killing.

5. The system of claim 4, further comprising one or more one-way valves interposed within the kill fluid line and operable to prevent the fluid flow in the passageway from entering the isolated kill tank.

6. The system of claim 1, wherein the sensor is an ultrasonic flow sensor.

7. The system of claim 1, wherein the controller comprises a memory, wherein the memory stores an algorithm for performing the well killing operation, data from the sensor, operational data of the HCR valve, or a combination thereof.

8. The system of claim 1, wherein the actuation mechanism is a hydraulic actuation mechanism operable to automatically open the gate in response to detection of the characteristic of the fluid flow in the passageway indicative of the need for a well kill operation.

9. A method comprising: detecting, with a sensor, a characteristic of a fluid flow through a passageway defined through a wellhead assembly, the wellhead assembly including a blowout preventer with one or more rams, and the sensor being mounted to an interior wall of the passageway; receiving, with a controller, a signal from the sensor indicative of the characteristic of the fluid flow; determining with the controller, and based on the signal, whether or not a predetermined set of conditions is met indicating a need for a well killing operation; sealing the passageway through the wellhead assembly by activating the one or more rams; receiving, with the controller, a closure signal indicating that the sealing of the passageway is complete, the closure signal being provided by at least one of the blowout preventer and the sensor; actuating, via an actuation mechanism controlled by the controller, a gate of one or more hydraulic control remote (HCR) valves within a kill fluid line extending from the passageway in response to receiving the closure signal; and activating one or more discharge pumps installed within the kill fluid line to pump a kill fluid to overcome a pressure of the fluid flow into the passageway through the one or more HCR valves.

10. The method of claim 9, wherein sealing the passageway through the wellhead assembly is performed prior to actuating the one or more HCR valves.

11. The method of claim 9, wherein activating the one or more discharge pumps is preceded by: mixing and preparing the kill fluid within a kill tank arranged adjacent the wellhead assembly; conveying the kill fluid to an isolated kill tank in fluid communication with the kill tank, the isolated kill tank being closed and sealed; and pressurizing the kill fluid within the isolated kill tank in preparation for the well killing operation.

12. The method of claim 9, wherein detecting the characteristic of the fluid flow includes operating an ultrasonic flow sensor.
13. The method of claim 12, further comprising disposing piping through the passageway, wherein the piping comprises drill pipe, production tubing, or coiled tubing, and wherein operating the ultrasonic flow sensor includes detecting the flow within the piping.
14. The method of claim 9, wherein the actuation mechanism is a hydraulic actuation mechanism, and wherein actuating the one or more HCR valves comprises hydraulically retracting the gate of the one or more HCR valves into a bonnet of the one or more HCR valves.
15. The method of claim 9, further comprising wirelessly communicating, via a wireless communication component of the controller, with the actuation mechanism, the sensor, and the one or more discharge pumps to remotely control operation thereof.
16. A system comprising: a wellhead assembly including a passageway extending therethrough and in fluid communication with a well; a sensor mounted to an interior wall of the passage and operable to detect a characteristic of a fluid flow in the passageway indicative of a need for a well killing operation; a blowout preventer comprising one or more rams operable to block and/or shear the passageway; a kill fluid line extending between a source of kill fluid and the passageway; one or more discharge pumps installed within the kill fluid line and operable to pump the kill fluid through the kill fluid lines; a hydraulic control remote (HCR) valve installed in the kill fluid line and including an actuation mechanism operable to remotely open and close a gate of the HCR valve; and a controller communicatively coupled to the sensor, the actuation mechanism, and the blowout preventer, wherein the controller is operable to remotely control the actuation mechanism, the one or more discharge pumps, and the one or more rams of the blowout preventer in response to detection of the characteristic of the fluid flow in the passageway indicative of the need for a well kill operation, and wherein the controller opens the gate after receiving a closure signal from the blowout preventer indicating actuation of the one or more rams.
17. The system of claim 16, wherein the actuation mechanism is a hydraulic actuation mechanism operable to raise and lower the gate of the HCR valve upon receiving a signal from the controller.